Quiz 1- Paper

January 15, 2026 1:59 PM

MATH 3042- Quiz 1 Full Name:
Time: 15 minutes Set:

1. (6 marks) This table shows the distribution of response times for 400 API requests to a particular web server.

Table: Response Time Distribution for API Requests (n=400)

Response Time (ms)	Frequency	Relative Frequency	Cumulative Relative Frequency
[0, 50)	143	0.3575	0.3575
[50, 100)	118	0.295	0.6525
[100,150)	76	0.19	0.8425
[150, 200)	38	0.095	0.9375
[200, 250)	19	0.0475	0.9850
[250, 300)	6	0.015	1.0000
Total	400	1	

$$Relative\ frequency = \frac{frequency}{n}$$

a) Find all missing values in the frequency column and the relative frequency column.
b) Is this distribution likely to be right tailed, left tailed or symmetric? Please explain your reasoning.

b) Likely a right-skewed distribution. The smaller x-values have higher frequencies, so the bulk of this distribution is clustered around $0 < x < 150$. As we move to right of x-axis towards larger x-values, there is a drop in the heigth of frequency columns which indicated the tail of this distribution.

2. (6 marks) A dataset containing daily cloud infrastructure costs for E-Commerce platform is summarized below:

$Min = \$2180$, $Max = \$8450$

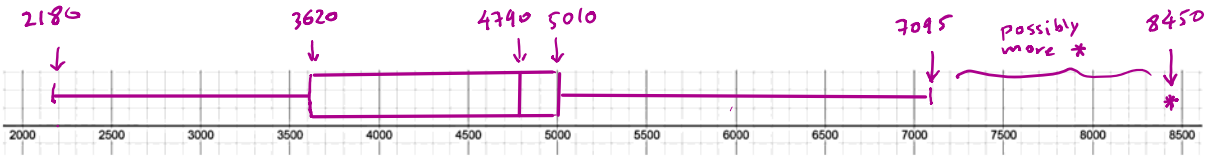
$Q_1 = \$3620$, $Med = \$4790$, $Q_3 = \$5010$

$IQR = 5010 - 3620 = 1390$
$UF = 5010 + 1.5(1390) = 7095$
$LF = 3620 - 1.5(1390) = 1535$

There are no lower outliers but there are at least one upper outlier

a) By calculating Upper and Lower fences, explain whether there exist(s) any outliers among the data.

b) Construct a boxplot of this data. If there are outliers put only one Asterix in the relevant end(s) of the boxplot representing all possible outliers. Use the provided number line to place values correctly (accurate within reason, I won't look for 100% accuracy of number placement) and draw the boxplot horizontally:



$IQR = Q_3 - Q_1$, $UF = Q_3 + 1.5 \times IQR$, $LF = Q_1 - 1.5 \times IQR$